BENJAMIN ATTAL

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EDUCATION

Carnegie Mellon Robotics Institute — *September 2019 - June 2025 (Expected)*
Ph.D. in Robotics — GPA: 4.0

Brown University — *September 2014 - May 2019*
B.S. in Applied Math and Computer Science, M.S. in Applied Math — GPA (within major): 3.9

RELEVANT WORK EXPERIENCE

CMU Light Transport Lab — *Fall 2019 - Present*
PhD Student with Matt O'Toole

Google Research — *Summer 2023*
PhD Student Researcher with Pratul Srinivasan

Facebook Computational Photography — *Summer 2021, 2022*
PhD Student Researcher with Changil Kim

Brown Visual Computing Lab — *Fall 2018 - Fall 2019*
Graduate Student Researcher with James Tompkin

Light — *Fall 2018 - Spring 2019*
Research Intern

AWARDS

Uber Fellowship — *2021*

Meta Fellowship — *2023-2025*
AR/VR Computer Graphics

CVPR 2025 Best Student Paper
Neural Inverse Rendering from Propagating Light

SELECTED PUBLICATIONS

Neural Inverse Rendering from Propagating Light. — **CVPR 2025 (Best Student Paper)**
Anagh Malik, Benjamin Attal*, Andrew Xie, Matt O'Toole, David Lindell.*

Flash Cache: Reducing Bias in Radiance Cache Based Inverse Rendering. — **ECCV 2024 (Oral)**
Benjamin Attal, Dor Verbin, Ben Mildenhall, Peter Hedman, Jon Barron, Matt O'Toole, Pratul Srinivasan.

Flowed Time of Flight Radiance Fields. — **ECCV 2024**
Mikhail Okunev, Marc Mapeke*, Benjamin Attal, Christian Richardt, Matt O'Toole, James Tompkin.*

Neural Fields for Structured Lighting. — **ICCV 2023**
Aarushi Shandilya, Benjamin Attal, Christian Richardt, James Tompkin, Matt O'Toole.

HyperReel: High-Fidelity 6-DoF Video with Ray-Conditioned Sampling. **CVPR 2023 (Highlight)**
Benjamin Attal, Jia-Bin Huang, Christian Richardt, Michael Zollhöfer, Johannes Kopf, Matthew O'Toole, Changil Kim.

Learning Neural Light Fields with Ray-Space Embedding Networks. **CVPR 2022**
Benjamin Attal, Jia-Bin Huang, Michael Zollhöfer, Johannes Kopf, Changil Kim.

Towards Mixed-State Coded Diffraction Imaging. **TPAMI 2022**
Benjamin Attal, Matt O'Toole.

TöRF: Time-of-Flight Radiance Fields for Dynamic Scene View Synthesis. **NeurIPS 2021**
Benjamin Attal, Eliot Laidlaw, Aaron Gokaslan, Changil Kim, Christian Richardt, James Tompkin, Matt O'Toole.

MatryODShka: Real-time 6DoF Video View Synthesis using Multi-Sphere Images. **ECCV 2020 (Oral)**
Benjamin Attal, Selena Ling, Aaron Gokaslan, Christian Richardt, James Tompkin.

TALKS

CVPR 2025 *Summer 2025*
Neural Inverse Rendering from Propagating Light

ECCV 2024 *Summer 2024*
Flash Cache: Reducing Bias in Radiance Cache Based Inverse Rendering

ICCP 2022 *Summer 2022*
Towards Mixed-State Coded Diffraction Imaging

Google (Invited) *Spring 2022*
Learning Neural Light Fields with Ray-Space Embedding Networks

ECCV 2020 *Summer 2020*
Real-time 6DoF Video View Synthesis using Multi-Sphere Images

SERVICE

Reviewer

- SIGGRAPH
- CVPR
- NeurIPS
- ICCV, ECCV
- ACM Transactions on Graphics
- Computer Graphics Forum

Organizer

- Workshop on Physics-inspired 3D Vision and Imaging (Pi3DVI) at CVPR 2025

TEACHING

Teaching Assistant

- Computer Vision (CMU 16385)
- Learning for 3D Vision (CMU 16889)
- Computer Graphics (Brown CSCI 1230)
- 2D Game Engine Development (Brown CSCI 1950N)

Head Teaching Assistant

- 3D Game Engine Development (Brown CSCI 1950U)